

# Getting the Lead Out

*Low-income families carry the heaviest burden of lead exposure. With the help of a chemistry class, that burden gets a little lighter.*



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**by Dennis C. Jacobs  
and John Casey**

**M**argaret and Jon Sauser were ecstatic when they bought their quaint, almost 100-year-old house in Kalamazoo, Michigan. They were overflowing with creative ideas on how to renovate the structure to make it the ideal home for their growing family. Jon started sanding and refinishing the hardwood floors, stripping the paint, removing some of the interior walls, and installing new windows. After six months of renovation, Jon, Margaret, and Jonnie, their two-year-old son, eagerly moved in. For reasons that they did not understand at first, Jonnie's behavior changed dramatically, and he was diagnosed with Attention Deficit Hyperactivity Disorder (ADHD). A short time later, their second son, Cameron, was born. At 11 months, Cameron stopped growing.

Knowing that something was tragically wrong, Margaret pleaded with her kid's skeptical pediatrician to test the children for lead. They discovered that both kids had been lead poisoned in the

"safety" of their own home. The amount of lead in each of the boys' bodies was  $2\frac{1}{2}$  times the action level set by the Centers for Disease Control and Prevention (CDC). Unbeknownst to Margaret and Jon, their dream home was plumbed with lead pipes and had been covered with lead-based paint. The Sauser's renovations had stirred up a significant amount of lead dust that would be slowly but steadily ingested by the family. Faced with mounting medical bills for the treatment of their sons' development problems, and with no funds for making their home lead safe, the Sausers had to abandon their home.

## *Lead in Homes, and in Children*

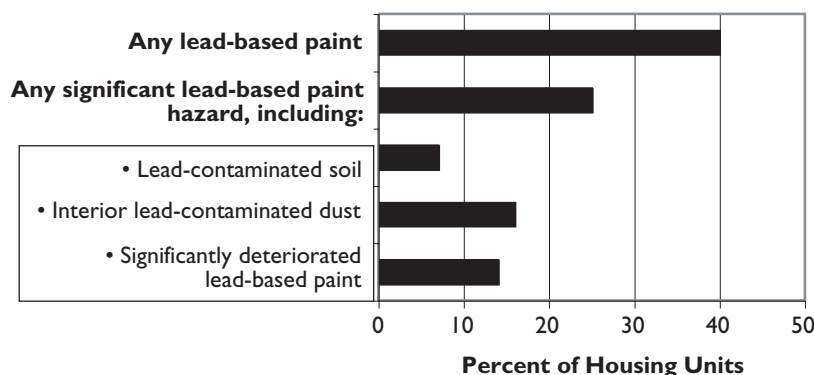
Lead poisoning presents a serious environmental health problem that is entirely preventable. The CDC reports that nearly one million children living in the United States have lead levels in their blood that are high enough to cause irreversible damage to their health. Lead poisoning most seriously affects children under the age of six, because it is precisely at this age that a

child's neurological function is developing most rapidly. Even low levels of lead are harmful and can be associated with reduced intelligence, impaired behavioral development, decreased stature and growth, and hearing loss. The CDC has set 10 micrograms per deciliter [ $\mu\text{g}/\text{dl}$ ] as the upper limit for blood lead levels in children, but recent studies suggest that one-third that level can cause cognitive decline.

In addition to removing lead from a child's environment, blood lead levels can be reduced through a healthy diet containing low-fat foods high in iron, calcium, and vitamin C. When the amount of lead in a symptomatic child's blood rises above  $40\text{ }\mu\text{g}/\text{dl}$ , immediate medical action, such as chelation therapy, is often warranted.

The symptoms of adult lead poisoning can include kidney and neurological damage, anemia, hypertension, impotence, sterility, and miscarriages. Although adults are less susceptible than children are to lead poisoning, pregnant women should be cautioned that lead could pass through the placenta and irreversibly harm a developing fetus.

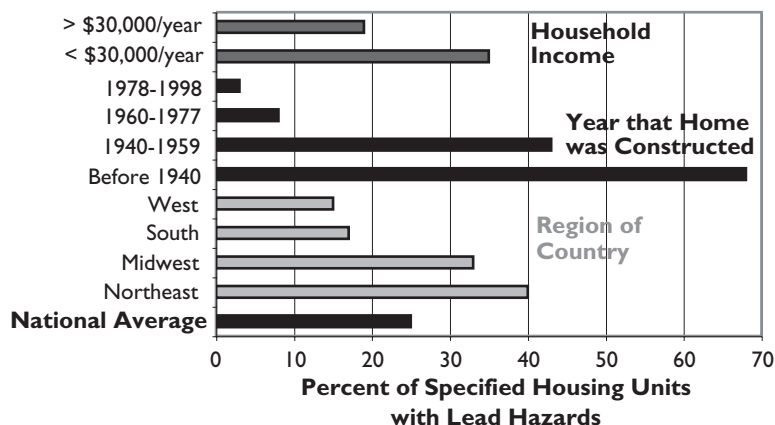
## Extent of Lead Hazards in the Nation's Housing Stock



Data is taken from "The Prevalence of Lead-based Paint Hazards in U.S. Housing" D.E. Jacobs, R. P. Clickner, J. Y. Zhou, S. M. Viet, D. A. Marker, J. W. Rogers, D. C. Zeldin, P. Broene, and W. Friedman, *Children's Health Articles* 101: A599-A606

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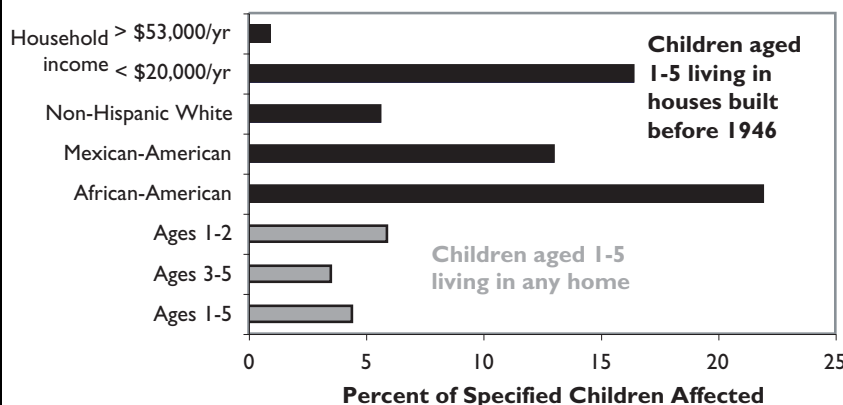
## Demographics of Homes Containing Lead Hazards



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## Children with Elevated Blood Lead Levels



Data is taken from the Centers for Disease Control

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## Keep at It

Lead levels that are currently assumed to be safe for children can significantly impair intellectual development, according to a report published in the *New England Journal of Medicine* in April. The current U.S. and international allowable blood lead level is 10 micrograms per deciliter (10 µg/dl), but researchers found that lead levels lower than that caused a drop in IQ of up to 7.4 points. That's sobering news, given that an estimated 1 in 50 U.S. children has lead levels above the federal standard, and 1 in 10 has levels of 5 µg/dl or higher—within the dangerous range, according to the report. Some 38 million houses across the country are still coated in lead-based paint, making homes the single largest source of the toxic substance.

—Doug Peckler

Doug Peckler is an associate editor of *Home Energy*.

The primary sources of lead in the home environment are deteriorating lead-based paint, lead-contaminated dust, and lead-infiltrated soil. Lead enters a child's body when the child either directly eats paint chips (lead gives paint a sweet flavor) or ingests lead-contaminated dust or soil through normal hand-to-mouth contact. Additional sources of lead-contaminated dust can be associated with a family member who pursues a hobby (such as making stained glass windows, bullets, or fishing sinkers) or an occupation that involves exposure to lead. Recognizing the serious health risks posed by lead, the Consumer Product Safety Commission banned lead as a component in residential paint in 1978.

A secondary source of lead in the home environment can be contaminated drinking water. Although lead pipes were commonly used in home construction before the early 1900s, it is rare to find lead fittings in homes built after 1970. Lead solder connections on copper pipes pose a much smaller risk. If a house contains lead pipes, the only way to completely

**Figure 1.** (top) About one-quarter of the houses in the United States have a hazardous amount of lead in its soil, dust, or flaking lead-based paint. **Figure 2.** (middle) Not every house is equal when it comes to lead hazards. **Figure 3.** (bottom) Poor children who live in houses built before 1946 face the greatest risk for lead exposure.

eliminate the risk of lead poisoning is to have the plumbing replaced. As a temporary measure, one can reduce the concentration of lead in potable water by running the cold water tap for 60 seconds before drinking from a faucet.

How prevalent is lead in the nation's housing stock today? An extensive HUD survey conducted in 1998–2000 revealed that approximately 38 million of the 96 million permanently occupied, noninstitutional housing units in the United States have lead-based paint. An estimated 29 billion ft<sup>2</sup> of painted exterior surfaces are currently covered with lead-based paint (see Figure 1). Although 40% of homes contain lead-based paint, as much as 26% of the nation's housing units pose a significant lead exposure hazard, according to current HUD/EPA standards. A home environment is considered hazardous if it contains one or more of the following: lead-contaminated soil, interior lead-contaminated dust, or significantly deteriorated or chipping lead-based paint.

Lead hazards can exist in any home, but data show that certain homes are more likely to pose a risk. The amount of lead in homes varies according to household income, the year the house was constructed, and the region of the



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Notre Dame students Erin Carmichael, Araceli Bonilla, Bob Stachecki, and Megan Farrell (left to right) join Professor Jacobs (center) in performing a risk assessment of a neighborhood home. After completing visual inspection of the exterior and interior, they collect paint, dust, and soil samples from many locations around the house.

country in which it is located (see Figure 2). It is clear that houses built before 1960 pose the greatest risk for dangerous lead exposure. In 1994, the CDC completed an extensive study examining the frequency of lead poisoning in children under the age of six. Poor children living in homes constructed before 1946 are at greatest risk (see Figure 3).

African-American and Mexican-American families are disproportionately affected by the health hazards of lead exposure, because they are more likely to be living in older homes and rental properties that have not been renovated to lead-safe standards.

## Paint Stabilization

When renovations are performed on an existing property that has not undergone a certified risk assessment, it must be assumed that all areas are positive for lead and that any disturbance of paint performed above the minimum amounts (2 ft<sup>2</sup> on any interior surface; 20 ft<sup>2</sup> on any exterior surface; or 10% of the total surface area of an interior or exterior component, such as a windowsill, baseboard, or trim) must be done using lead-safe work practices.

The following procedure constitutes a HUD-approved strategy for stabilizing lead-based paint. Use appropriate safety and personal protective equipment or procedures as outlined by federal and state laws in your area. Before deteriorating lead-based paint is removed or resurfaced, the affected area must be

precleaned. First, clean any removable materials (such as light fixtures or hardware), enclose them in plastic, and remove them from the work area. Second, preclean the entire room, including the ceiling, walls, floor, and all fixed building materials. Precleaning consists of vacuuming all surfaces, beginning with the ceiling and moving in a downward motion on each wall, using a vacuum cleaner specially equipped with a HEPA filter.

Next, wipe down all vacuumed surfaces with a wet cloth soaked in mild detergent. Trisodium phosphate (TSP) has often been recommended for the wipe-down step, because it binds tightly with lead to form a precipitate—lead phosphate. However, TSP presents health hazards itself and should be handled with caution. After the ceilings and walls have been wiped down, rinse all the washed

areas twice. Next, vacuum the treated surfaces once again with a HEPA filter-equipped vacuum. The wash and rinse cycles should always be performed with separate buckets, never exchanging rags or water between the buckets. Replace the rags frequently and dispose of the used rags; do not attempt to clean or reuse them.

This procedure should be executed directionally in each room or affected area, starting at the far corner and ending at the entrance to the room. Within 1 to 24 hours after the cleaning procedure is completed, the flooring in each treated room must be sealed with wax, polyurethane, concrete sealer, or another appropriate sealer. After the sealer has dried, a layer of 4–6-mil plastic should be attached to the floor and 6 to 12 inches up each wall, to cover the cleaned area

## Making Homes Safe

What can be done to identify and minimize the risk of lead exposure? In St. Joseph County, Indiana, a collection of nonprofit organizations, government agencies, and private companies have banded together to form a Get the Lead Out task force. The coalition works to inform the public about the dangers of lead, provides free blood screening for young children, and performs risk assessments in homes where children have tested positive for lead exposure. Through the Women, Infants, and Children (WIC) program administered through Memorial Hospital, nearly 2,000 children in the county are tested for lead each year. Approximately 6% of the children test positive for having blood lead levels above the CDC action limit of 10 µg/dl.

The University of Notre Dame has recently formed a unique partnership with member organizations in the Get the Lead Out task force. Advanced undergraduates can enroll in a course entitled Chemistry in Service of the Community and receive academic credit for their participation in the program. Students disseminate information on lead poisoning to low-income families living near campus and encourage parents to have their children tested for potential lead exposure. A portion of the

families that Memorial Hospital identifies as having elevated lead levels are referred to the Notre Dame class for a detailed lead risk assessment. A risk assessment involves visual inspection of the home; collection of dust, paint, and soil samples to be analyzed later in one of Notre Dame's chemistry labs; and on-site measurement of lead using an X-ray fluorescence (XRF) spectrometer. The spectrometer will detect the presence of lead on a wall, even if the lead is buried below many layers of nonlead-based paint.

What constitutes a significant level of lead contamination? Federal guidelines define a lead paint hazard as deteriorated paint containing more than 650 µg of lead per ft<sup>2</sup> of painted area. The clearance standards for lead-contaminated dust are 40, 250, and 400 micrograms of lead per square foot for floors, windowsills, and window troughs, respectively. The lead hazard standards for soil are 400 ppm by weight in play areas and 1,200 ppm in other areas of the yard. State and local regulations can be more stringent than these federal guidelines, so check with local ordinances to find the action levels in your area.

Approximately 80% of homes tested by Greentree Environmental Services, a member of the Get the Lead out task

force, test positive for lead. In a recent risk assessment, the team of students and professionals discovered that a rented apartment occupied by a family with four young children had dangerously high levels of lead on the tread of the entrance stairwell, the painted front porch, and the casework around the entrance to the apartment building. Because these were areas frequented by the children, it was no surprise that during normal play activity, the kids were ingesting lead-contaminated dust that they picked up on their hands. In fact, prior blood tests had revealed that the two-year old daughter had a blood lead level equal to 26 µg/dl, whereas her older sibling, who did not crawl around on the floor, had a lead level of just 3 µg/dl.

After completing laboratory analysis of the dust and soil samples, the team of students and professionals returned to the home to share their findings with the tenants and the landlord. They discussed inexpensive ways to minimize the amount of lead-contaminated dust in the home. These included dusting regularly with a wet cloth or using a vacuum cleaner equipped with a high-efficiency particulate air (HEPA) filter. They also presented a longer-term solution—paint stabilization. The cost for paint stabiliza-

and keep it clean until the job is completed and ready for clearance inspection.

Carpeted floors require a more complex cleaning procedure. If a child will come in contact with area rugs, dispose of them. It is widely believed that these rugs can never be completely decontaminated. For cleaning, use a vacuum with a beater bar for one minute per square yard of surface area. The rug or carpet should then be folded and the underside vacuumed without using any beater bar. Flip the folded rug or carpet over and repeat on the opposite side. The uncovered floor should be vacuumed while it is exposed. Finally, the top surface of the rug or carpet should be revacuumed at the rate of two minutes per square yard.

It is futile to try to totally remove all the lead from wall-to-wall carpeting. If replacement is not possible, the cleaning process recommended is four minutes of vacuuming

per square yard—two minutes in one direction and two minutes in a perpendicular direction. Then the carpeting should be steam cleaned.

Porous furniture should be vacuumed for at least one minute per square yard. It is recommended that ductwork, as well as vent covers and cold-air return grates also be cleaned prior to clearance.

For house exteriors, an initial visual inspection should concentrate on the drip line of the home. The drip line is the area 5 to 7 ft out from the foundation, or the point of the house where the roofline ends as an overhang from the structure. Look for any visible paint chips or any area of bare soil that exceeds 9 ft<sup>2</sup>. Any visible paint chips should be HEPA vacuumed up. The same thing should be done around any outbuildings on the property that a child might come in contact with.

Plastic should then be placed 10 ft out from the affected area in all directions (when possible), plus an additional 10 ft for each story on which the work is to be performed. Any surface preparation work should be done wet—that is, the surface should be misted first and then scraped. The debris should be HEPA vacuumed and wet wiped. Then the plastic should be folded back into itself, duct taped, and bagged, or placed in a dumpster or appropriate container. Areas of bare soil should then be covered with at least 6 inches of mulch or gravel and monitored as prescribed in the risk assessment report. (A licensed risk assessor would normally perform a risk assessment on the property if federal money of over \$5,000 were used there.) Usually this means monitoring every 12 to 18 months by the risk assessor.



JOHN CASEY

When stabilizing paint on a home's exterior, plastic should be placed at least 10 ft out from the wall.

tion by a qualified contractor runs between \$3,000 and \$5,000 for an average-sized home, depending on the extent of contamination (see "Paint Stabilization").

### Training for Lead Safety

What does it take to become a qualified contractor for lead-safe renovations? The new HUD regulations on lead hazard reduction requirements were released in September 1999. The Lead Safe Housing Rule places increased emphasis on reducing the risk of exposure to lead through dust, because this is the primary way that young children become lead poisoned. HUD has established two high-quality training courses—one for landlords and one for contractors. Many participants in the courses are initially apprehensive about "the government trying to tell us how to do our jobs." For the first 30 minutes, one often hears attendees saying, "This lead-paint thing is a bunch of bull. I have been doing this for 20 years and never had a problem." Then their anger subsides, and they begin to think of their own children and whether they might have been exposed to lead risks.

Their complaints are transformed into questions. "Why aren't doctors trained to identify lead poisoning? My child is on Ritalin for hyperactivity. Do you think that he might be lead poisoned?"

The remodeling and weatherization industries continually seek out innovations to improve energy conservation, safety, or efficiency. In the same way that we expect our doctors to stay up-to-date with the latest technology or medical treatments, homeowners justifiably expect that their contractors will use lead-safe practices when weatherizing or remodeling a house (see "Lead-Safe Weatherization," *HE* May/June '03, p. 24). Homeowners have a right to receive a disclosure statement from their contractor informing them that lead-based paint may be disturbed during a job. They have a right to expect that this hazard will be contained, controlled, and removed upon job completion. This is the essence of HUD's Lead Safe Housing Rule.

Lead-safe practices include containing dust by laying down 6-mil plastic, 5 ft outside the work area on the interior and 10 ft outside the work area on the exterior, before you start working. Wet sanding must be used rather than

conventional dry-sanding techniques. Maintaining a clean job site reduces injury, saves cleanup time at the end of the job, and eliminates complaints from the homeowner. Showing care and concern for your client's family will win you a lifelong customer, and it can lead to valuable referrals.

Contractors, landlords, and tenants sometimes resist attempts to educate them about the dangers of lead. If you are a skeptic, sit down with the parents of a lead-poisoned child and listen to their story. If you are still not convinced, speak to your attorney about the potential liability you face with regard to litigation by homeowners or your workers or both. Renovations conducted in the absence of lead-safe practices needlessly put many people at risk for contracting serious long-term health problems.

In 2000, the President's Task Force on Environmental Health Risks and Safety Risks to Children outlined a plan to virtually end childhood lead poisoning in America within ten years. The report estimates that by preventing the adverse effects of lead on children's health and development, the long-term economic benefits will exceed the cost of the strategy by \$8.9 billion. We can't afford not to act.



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### For more information:

To locate qualified lead-safe contractors in your area or to learn where you can be trained in lead-safe practices, consult the National Lead Service Provider's Listing System, [www.leadlisting.org](http://www.leadlisting.org).

HUD posts a useful and comprehensive Web site, [www.hud.gov/offices/lead](http://www.hud.gov/offices/lead), on healthy homes and lead hazard control.